CarbonTrace AI

AI-Powered Export Compliance for MSMEs

Small exporter. Enterprise-grade compliance.

TEAM NAME: ACERE

THE PROBLEM

MSMEs Are Not Losing on Product Quality : They Are Losing on Evidence Chaos

EUDR Pressure

Geolocation and deforestation-risk proof required for relevant commodities entering the EU market.

CBAM Pressure

Installation, fuel, electricity, and production emissions data required for covered goods from 2026.

Lost Orders

Importers shift to larger suppliers when MSMEs cannot provide structured evidence quickly.

€120B

Goods trade volume in 2024 (European Commission).

45.7%

Of Indian exports in 2023–24 (PIB/MSME sources).

2026

Definitive regime applies after the transitional period.

⚠ The real gap is not product supply — it is trusted proof supply.

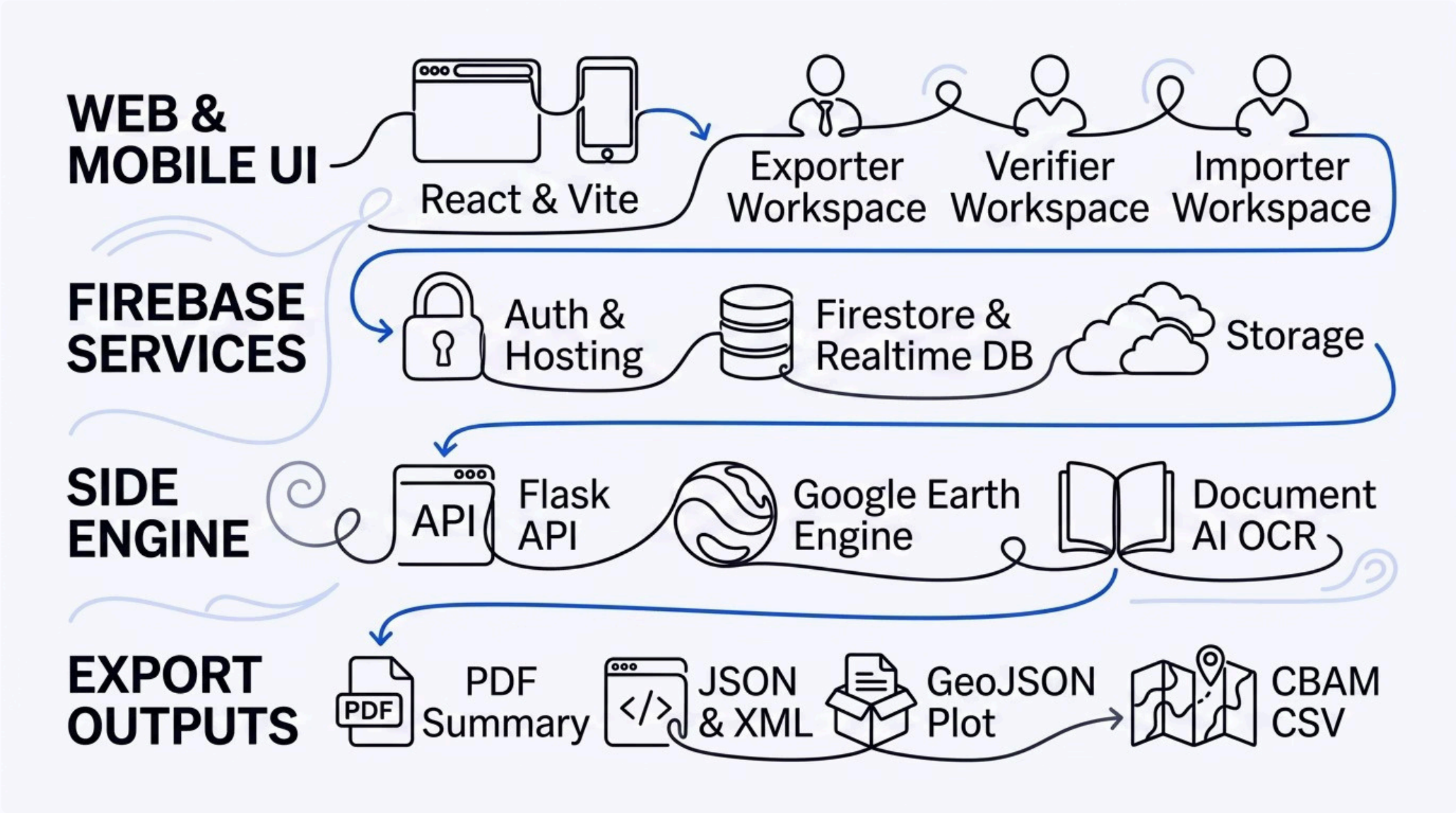
THE SOLUTION

One Connected Compliance Workspace : From Exporter Data Entry to Importer Handoff

- CarbonTrace AI is a complete shipment evidence workflow—not a formilling tool. Exporters build structured records, approve linked evidence, and importers receive only verified packages.
- Exporter creates linked records: profile, suppliers, plots, documents, facilities, batches, and shipment.
- EUDR analysis and extraction review turn raw evidence into structured compliance data.
- Verifier reviews one connected case instead of scattered files and emails.
- Importer receives only approved packages with PDF, XML, JSON, CSV, and GeoJSON outputs.
- Scattered files become one trusted shipment package.

TECHNICAL APPROACH

Firebase for Infrastructure, Flask + Google Earth Engine for Geospatial Intelligence



Frontend
React + Vite with role-based workspaces

Backend
Flask API + Google Earth Engine

Infrastructure
Firebase Auth, Firestore, Storage, Hosting

Exports
PDF, JSON, XML, GeoJSON, CBAM CSV

AI accelerates the workflow, but reviewed evidence creates trust.

FEASIBILITY & VIABILITY

Start With One High-Quality Shipment Workflow, Then Scale by Sector and Buyer Networks

Our MVP covers exporter profiles, suppliers, plots, document uploads, extraction review, facilities, production, shipments, verifier queues, and importer packages — one perfect flow before scaling.

1

Build Now

Exporter workspace, plot mapping, document review, verifier queue, importer download.

2

Pilot Next

Onboard 5–10 exporters in one sector cluster (coffee or industrial). Process real shipments. Collect feedback.

3

Scale Later

ERP/Tally imports, multilingual onboarding, buyer dashboards, and association-level cluster packages.

SaaS Subscription

Starter and Growth plans for exporters

Shipment Credits

Pay-per-package for active exporters

OCR & Report Add-ons

Premium extraction and compliance reports

Enterprise Dashboard

Importer-side supplier monitoring

One solved shipment becomes a repeatable compliance product.

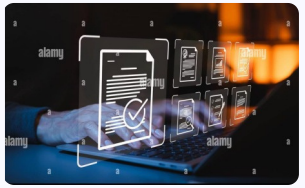
IMPACT & BENEFITS

Faster Readiness, Better Verification, Cleaner Importer Trust, Stronger MSME Market Access

CarbonTrace AI creates multi-sided value—exporters save time,verifiers decide faster, importers gain confidence, and MSMEs compete with enterprise-grade compliance.

Exporter

Less document chaos, faster package creation, reusable profiles, and stronger buyer confidence.



Verifier

Structured queue with linked evidence, side-by-side review, and fewer email follow-ups.

Importer

Approved packages, key documents, compliance summaries, and downloadable artifacts.

MSME Ecosystem

Smaller exporters meet modern compliance expectations without enterprise compliance teams.

Better evidence creates better market access.

RESEARCH & REFERENCE

Regulation, Trade Volume, and MSME Pressure Prove This Problem Is Real

CarbonTrace AI is grounded in real regulatory pressure and verified trade data—not an invented problem. Every feature maps to alive compliance requirement or documented market need.

EUDR — EU Deforestation Regulation

Requires due diligence and geolocation data for relevant commodities. CarbonTrace AI maps plots and runs geospatial analysis directly in the export workflow.

CBAM — Carbon Border Adjustment Mechanism

Definitive regime from 2026 requires emissions data for covered sectors. CarbonTrace AI captures installation, fuel, and production inputs at the batch level.

EU-India Trade Corridor

Goods trade reached ~€120 billion in 2024 (European Commission). MSMEs contribute 45.7% of Indian exports — a massive compliance-ready market.

Technology Feasibility

Google Earth Engine supports large-scale geospatial analysis. Firebase provides managed auth, database, storage, and hosting for rapid deployment.

The project is grounded in regulation, trade, MSME economics, and feasible technology.

TARGET AUDIENCE & UNIQUENESS

Designed for MSME Exporters Who Need Buyer-Trust Proof : Not Another Generic Document Folder

CarbonTrace AI targets MSME exporters under EU compliance pressure, with verifiers, sustainability consultants, and importers as secondary users. Beachhead sectors: coffee and agro-commodities (EUDR) and steel and aluminium (CBAM).

Connected Evidence Graph

Suppliers, plots, documents, facilities, and shipments linked around each shipment — not just file storage.

Plot-Level EUDR Analysis

Geospatial intelligence integrated directly into the export workflow — not a separate tool.

Verifier + Importer Trust Loop

Verifier approval and importer download complete the chain from source data to buyer handoff.

MSME-First Design

Guided workflows, reusable compliance records, and low-friction onboarding built for small exporters.

> CarbonTrace AI helps MSMEs prove — not just claim — that their shipment is ready for global buyers.
> Scattered evidence → Verified package → Global buyer trust.